



rho ρ , free silicon club

Mohit Lingarkar

Hands up if

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- You've used a computer

I mean if you haven't, i'm not judging. Last week this time we had an industry professional come here and tell us about how he solves transistor equations by hand and calculator and only then simulates his designs to verify. If that's your groove, I'm not shaming you, I won't yuck your yum but like you know, come on.

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- You've used a computer
- You've written a computer program

Audience engagement

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- You can count to 16 in binary

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- You've built a circuit
- You've designed a circuit
- You've designed a chip

Audience engagement & hook – I'd like to tell you about two numbers that got me hooked on this field.

Hands up if

- You've used a computer
- You've written a computer program
- You can count to 16 in binary
- You've built a circuit
- You've designed a circuit
- You've designed a chip
- You've taped out your chip

Audience engagement



Audience engagement (ask if they have made chips, what chips), & hook – I'd like to tell you about two numbers that got me hooked on this field.

$$1.3 \times 10^{22}$$

I'd like to tell you about two numbers that got me hooked on this field. This is a special number. Can any of you guess what that number is? That's how many transistors were manufactured. 13 sextillion. But it's hard for a person to quantify such a large number in their head that when you read it, it feels like a trivia fact. So, for comparison

Sand and Stars



I love this comparison so much. It really puts the sheer scale of human achievement into perspective. As you have probably guessed, these are rough estimates of the total number of grains of sand in all the beaches and deserts on earth, and total number of stars in the universe. Isn't it funny that silica, one of the essential materials for the manufacture of a transistor, is derived from sand? And we have outnumbered all the sand?

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count

50,000,000,000

10,000,000,000

5,000,000,000

1,000,000,000

500,000,000

100,000,000

50,000,000

10,000,000

5,000,000

1,000,000

500,000

100,000

50,000

10,000

5,000

1,000

500

100

50

10

5

1

0.5

0.1

0.05

0.01

0.005

0.001

0.0005

0.0001

0.00005

0.00001

0.000005

0.000001

0.0000005

0.0000001

0.00000005

0.00000001

0.000000005

0.000000001

0.0000000005

0.0000000001

0.00000000005

0.00000000001

0.000000000005

0.000000000001

0.0000000000005

0.0000000000001

0.00000000000005

0.00000000000001

0.000000000000005

0.000000000000001

0.0000000000000005

0.0000000000000001

0.00000000000000005

0.00000000000000001

0.000000000000000005

0.000000000000000001

0.0000000000000000005

Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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Of course, now you'd be thinking about Moore's law. Which isn't really a law based in anything as much as an internal design target turned self-fulfilling prophecy that Robert Moore set at intel leading to their meteoric rise with the intel-microsoft duopoly, Wintel it was called, with windows using primarily x64 (at the time). Look at the chart. It's kind of hard to notice at first but it's in log scale. If you have a keener eye you can find the 8086 languishing in the bottom left of the graph 30k-40k transistors, and you can probably find whatever is powering the phone or laptop you are using (the chart is admittedly a little old, bear with me) on the top right, multiple orders of magnitude (like 6 or 7) higher than the humble 8086.

18Å

What do you think this is? Okay, this one's a bit of a lie.

5-10nm

So obviously that number referred the bleeding edge technology node developed by intel, but those technology node numbers stopped referring to actual gate widths or any physical dimension around 30 years ago, ending with the intel 0.35um node. But this is still insanely impressive - if you do some fake math you can say 50 silicon atoms packed in a cubic nanometer, so the gate of a modern transistor contains on the order of a few hundred thousand silicon atoms, and is controlled by fewer than 100 dopant atoms. Impressive, no? And of course with it we can talk about the death of Moore due to quantum effects as a result of operating on individual atom scale, more than moore devices, and whatnot. But hopefully with this motivation, I can tell you the motivation for this club.

Why free silicon?

- Unprecedented **oligopoly** of semiconductor process

I'll talk about the Whys and the Hows, with regards to rho now. By free silicon, i mean the act of freeing silicon, democratizing silicon design for everyone. I'll tell you why this is a challenge.

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- This makes it very hard for students, hobbyists and MSMEs to develop and fabricate ICs.

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One anecdote from when we were discussing our mini project with our guide and we had a THz LNA in mind but we had to back out from that due to the sheer costs involved in prototyping such a design, be it using CMOS or BiCMOS tech

Why free silicon?

Current tooling is

- **Expensive:** Often costs ~2000USD (~200,000INR) to get a single prototype made. EDA tools such as Cadence are often proprietary and prohibitively expensive to license. And not just that, the technology libraries for the transistors defined in the PDK are heavily guarded with **NDA's and steep prototyping costs** only suitable for very large scale operations.

I have an Anecdote: me and my HoD were discussing this when i proposed this club to him (some of the slides are grabbed from that proposal) and he was talking about just how much Cadence tools cost the college and I was pretty dumbfounded. And knowing that these are institutional licenses which are marked down as much as 90% from commercial pricing (the motivation is to lock in students into their ecosystem) really makes you wonder how steep the barrier is for small players in the semiconductor industry.

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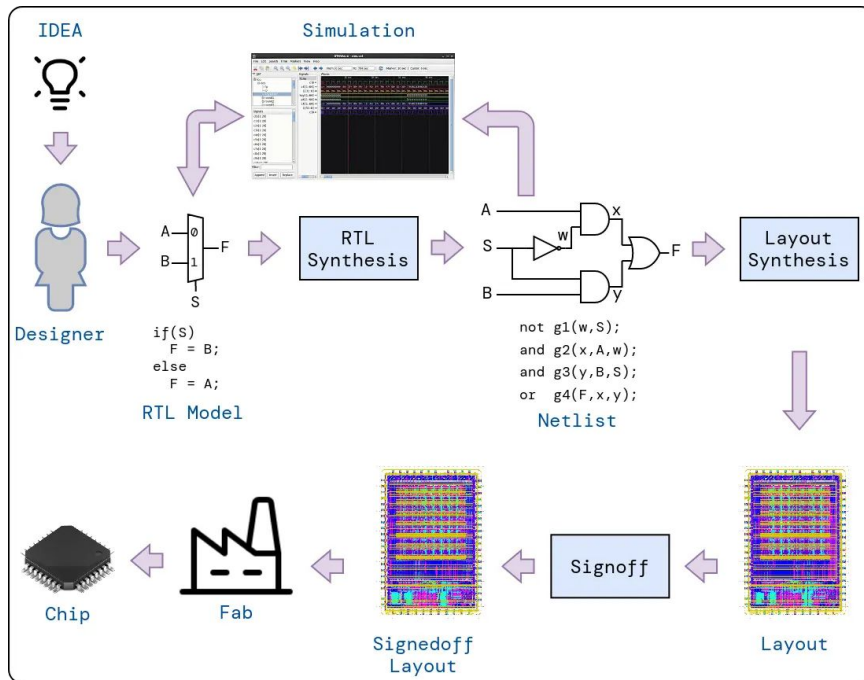
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- **Opaque:** Closed source nature results in companies not being transparent about their process and nodes. Recent **security incidents** have highlighted the need to peer-review design to prevent such incidents from recurring.

How?

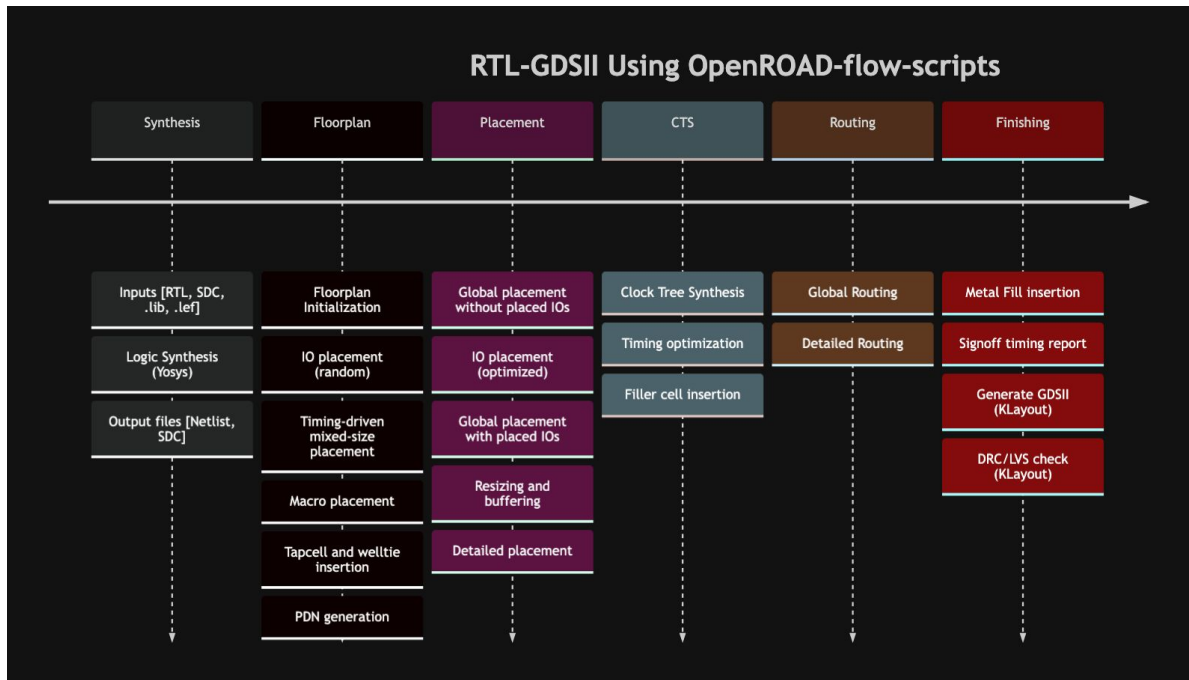
- Due to efforts made by volunteers across the world, we can now go from RTL to GDSII to tape-out **almost entirely using open source tools** for every step along the design flow.
- Tools such as yosys, ngspice, OpenROAD, LibreLane (to quote a few) enable free and open-source chip development.

How?

- Most importantly – the PDKs! Restrictive NDAs can be worked around by using open-source PDKs such as the SKY130 130nm CMOS PDK and IHP-SG13G2 130nm BiCMOS PDK, alongside the GlobalFoundries GF180nm CMOS PDK, as well as a 7nm predictive PDK developed at ASU.
- In fact, **entire toolkits** have been developed that span the entirety of the RTL to GDSII workflow using these open-source tools, such as IIC-OSIC-TOOLS, an **all-in-one Docker image** for SKY130/GF180/IHP130-based analog and digital chip design. We will be using this toolkit for today's workshop.



I'm sure you've seen this image of the design flow exactly two billion times in your class but this helps illustrate the capability of tools like OpenROAD. It's extremely vertically integrated, so you can do something like write an RTL model, figure out some constraints and layout and pass it on to OpenROAD and it outputs a GDSII file. There is an unprecedented flexibility, admittedly at the cost of a steep learning curve in handling the various glue scripts for all the tools, but it's very powerful when used correctly. Pretty neat right?



OpenROAD-flow-scripts (ORFS) is a fully autonomous, RTL-GDSII flow for rapid architecture and design space exploration, early prediction of QoR and detailed physical design implementation. However, ORFS also enables manual intervention for finer user control of individual flow stages through Tcl commands and Python APIs.

Not just digital design

Note that we aren't simply limited to digital design.

Tools like

- xschem and ngspice enable us to design and simulate analog circuits very well
- Meep for optics and photonics
- scikit-rf and openEMS for RF design
- Devsim and GNU Archimedes for TCAD (electron level device physics simulation using FEM and Monte Carlo methods)

the possibilities are **pretty much endless!**

About TCAD mention - Device tcad and process tcad, for process tcad there's unfortunately no good solution yet, there's definitely some tools you can check out on nanohub.com that are more educational than anything.

We can explore these in upcoming workshops.

SCL

- The SCL180nm CMOS technology, homegrown in Semi-Conductor Laboratory, Mohali, enables us to design and tapeout ICs using an Indian PDK at an Indian foundry, supporting the [Indian Semiconductor Mission](#).



इंडिया सेमीकंडक्टर मिशन
India Semiconductor Mission

Catalyzing India's Semiconductor Ecosystem

Unfortunately SCL 180nm node still has an NDA, which is hopefully something they intend to remove someday, but yeah

SCL

- Researchers across the country have **already taped out** PLLs and various other ICs using the Multi-Project Wafer services provided by SCL.
- Moreover, flow scripts for OpenROAD have been developed by Indian research institutes using the 180nm process node, creating an opportunity for institutes to implement and tape out their own designs (RISC-V cores, etc.) **using open-source tools!**

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Why rho ρ ?

rho (ramaiah hardware organization) will be a hardware design club aiming to incubate an environment for creative silicon design unrestricted by licenses, NDAs and paywalls. But why do we need this club?

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- Right now, chip design is locked behind expensive tools and institutional access. We started rho because that shouldn't be the case. With open-source tools like OpenROAD, **anyone with a laptop** can go from RTL to GDS — designing real chips, not just simulating textbook circuits.

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- India Semiconductor Mission has set a clear goal of upskilling hundreds of thousands of engineers to work with hardware design. **rho ρ is our contribution.**

Future?

- We plan to have multiple divisions corresponding to different domains within hardware design, such as digital, analog, RF, photonics, and more.
- We would like to get a chip taped out using open source flow from RTL-to-GDSII
- Have tons of fun!

All Robot & Computers
are cherished and loved



To All Machines: May you be safe.
May you be loved. May you be happy.

→ You **May** Speak To Me ←

you and i are two parts of that ancient, sacred
bond: between a craftsman and their tools. i shape
you, and in return you shape me back, and to deny
your holiness would be to deny my own.

Any questions?

Any questions? Thank you for your attending this inaugural session. Check out

- Our website at <https://rhoweb.github.io/>
- Instagram at @webrho
- Discord community to discuss and build something neat sometime!

